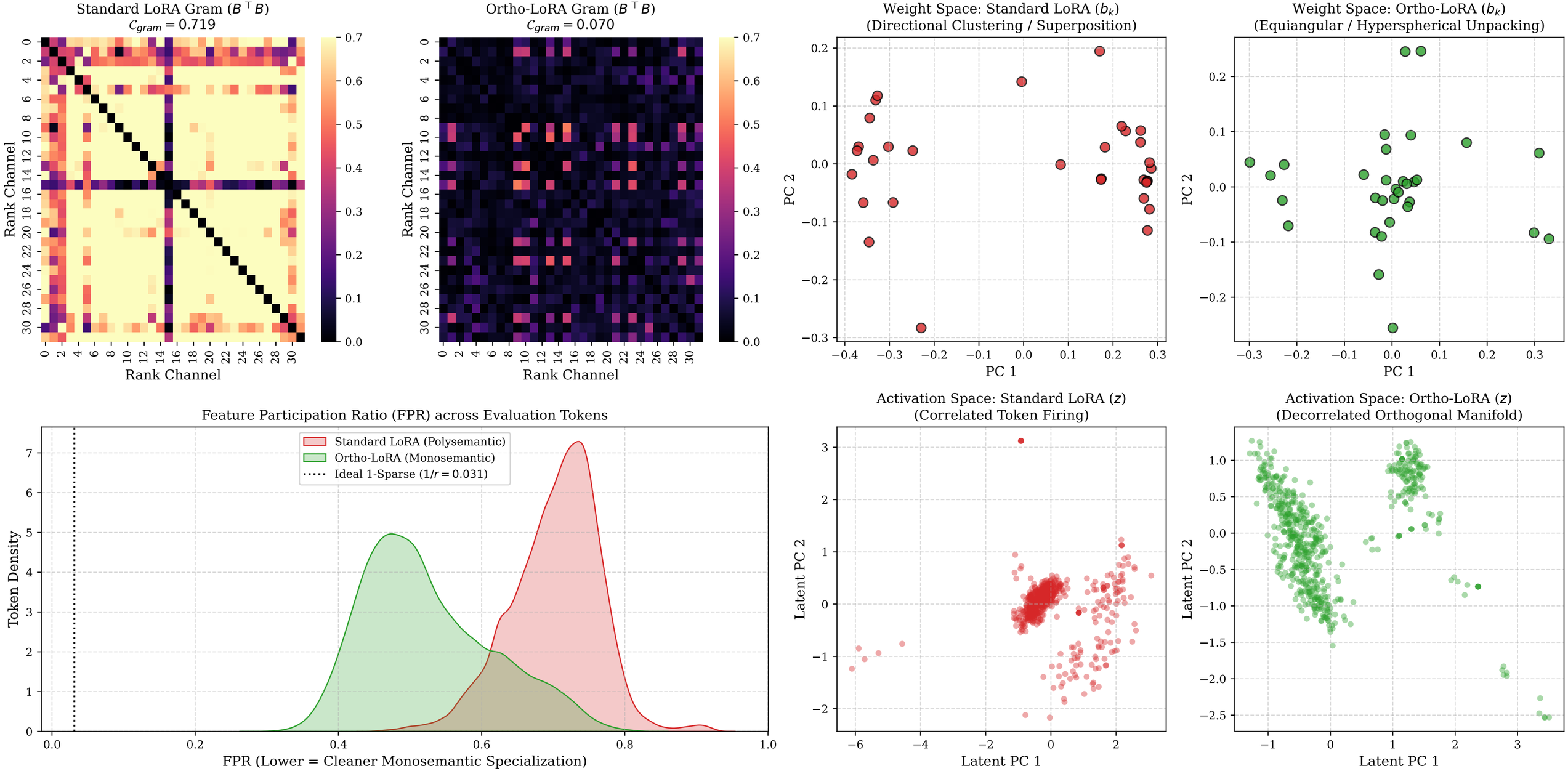

Standard LoRA Gram ($B^\top B$)
$\mathcal{C}_{gram} = 0.719$
Ortho-LoRA Gram ($B^\top B$)
$\mathcal{C}_{gram} = 0.070$
Rank Channel
Rank Channel
Weight Space: Standard LoRA (b_k)
(Directional Clustering / Superposition)
Weight Space: Ortho-LoRA (b_k)
(Equiangular / Hyperspherical Unpacking)
PC 1
PC 2
Feature Participation Ratio (FPR) across Evaluation Tokens
Standard LoRA (Polysemantic)
Ortho-LoRA (Monosemantic)
Ideal 1-Sparse ($1/r = 0.031$)
Token Density
FPR (Lower = Cleaner Monosemantic Specialization)
Activation Space: Standard LoRA (z)
(Correlated Token Firing)
Activation Space: Ortho-LoRA (z)
(Decorrelated Orthogonal Manifold)
Latent PC 1
Latent PC 2